

pay as per usage

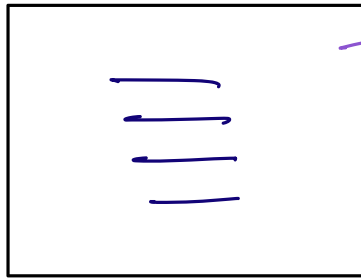
→ Auto Scaling

del.icio.us

DHCP.

↓
Dynamic Host
Configuration
Protocol.

Laptop



→ Public Ip + Static Ip

del.icio.us

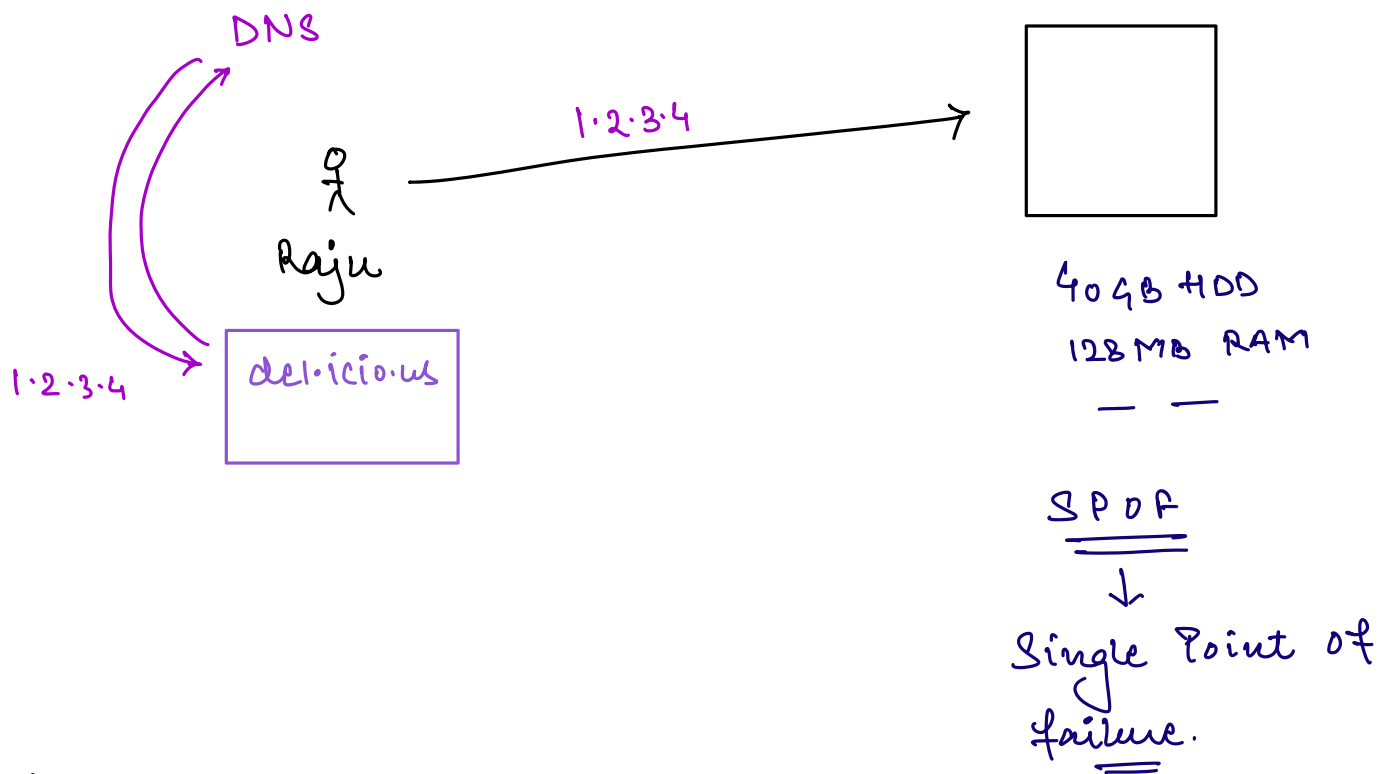


~~1.2.3.4~~

7.8.10.11

→ Bookmarking
website.

MUP



Scale Estimation

↳ Back of the Envelope Calculations.
(Question)

users

id	email	password
8B	30B	30B

↘
~ 100B.

bookmarks

—	—
8B	—
url	—

→ 1KB

1M users $\Rightarrow$ 100 Bytes $\times$ 1M
 $\Rightarrow$ 100MB.

1 M Bookmarks / Day $\Rightarrow 1M \times 1KB.$
 $\Rightarrow \underline{1GB. / Day}$
 1000 B

Bit

1 Byte

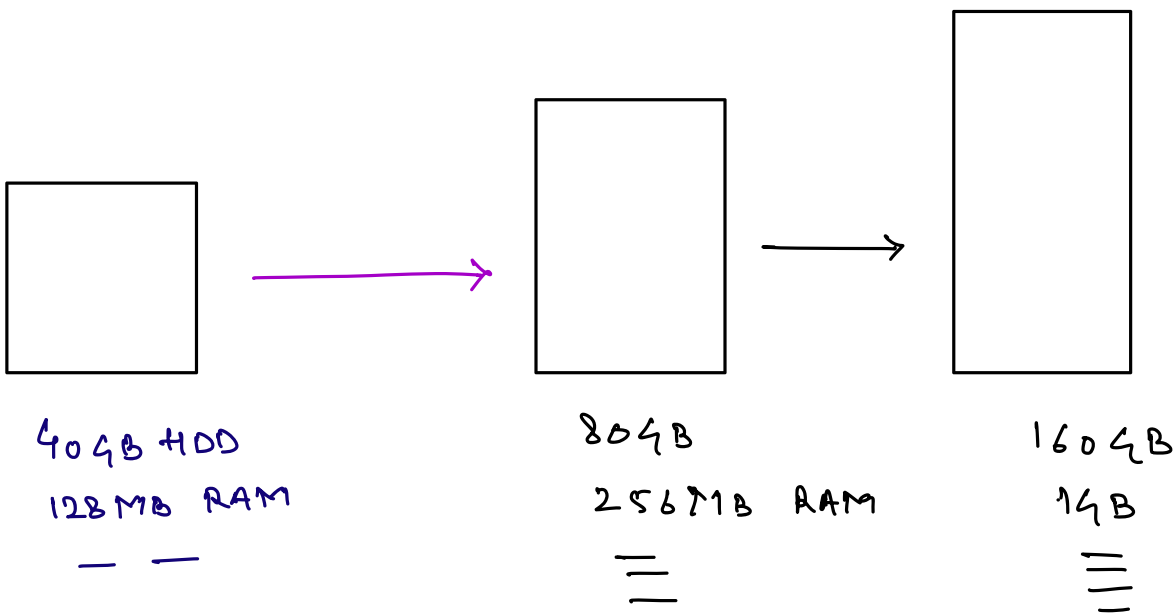
1000 Bytes = 1KB

1000 KB = 1MB

1000 MB = 1GB

1000 GB = 1TB

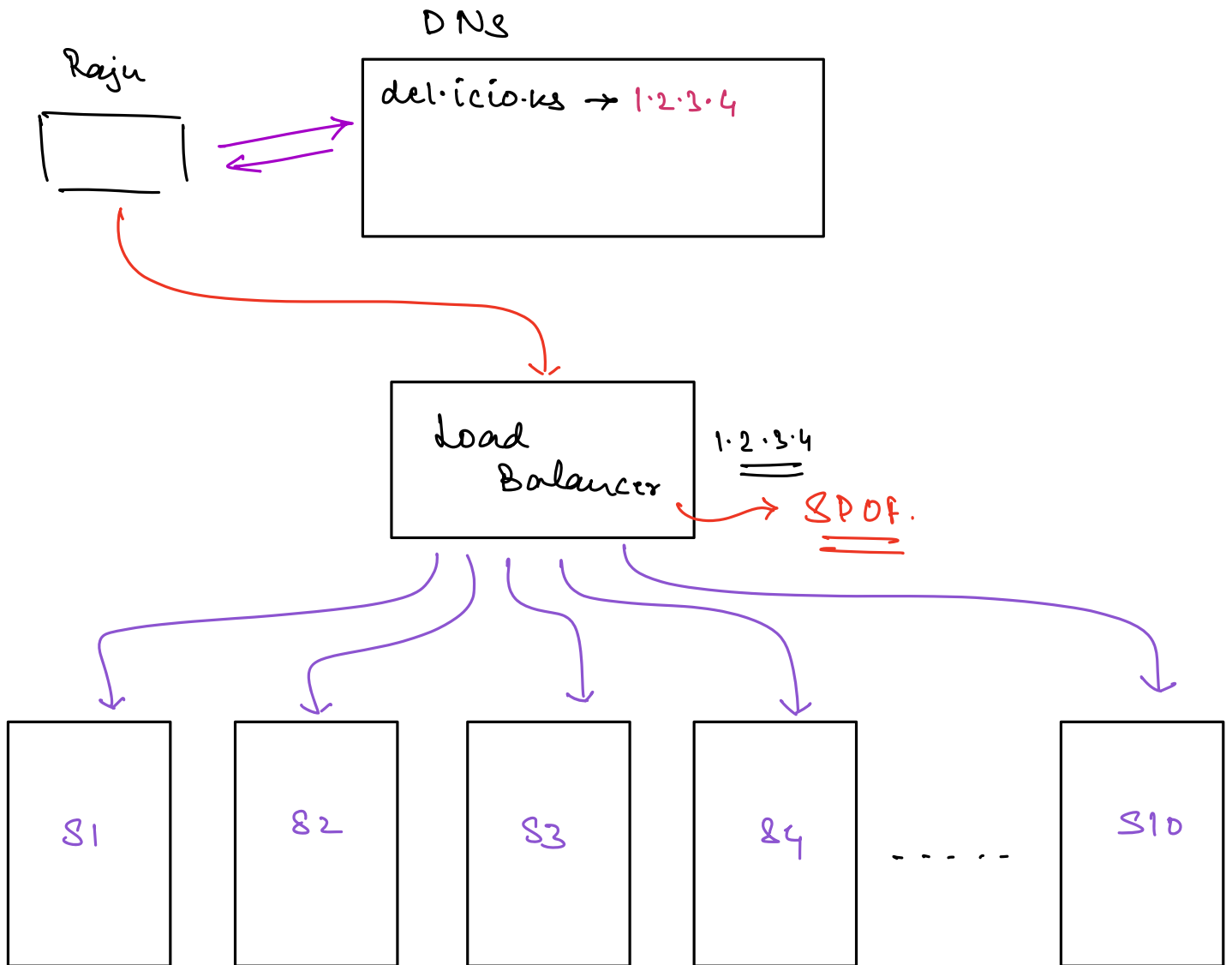
1000 TB = 1PB

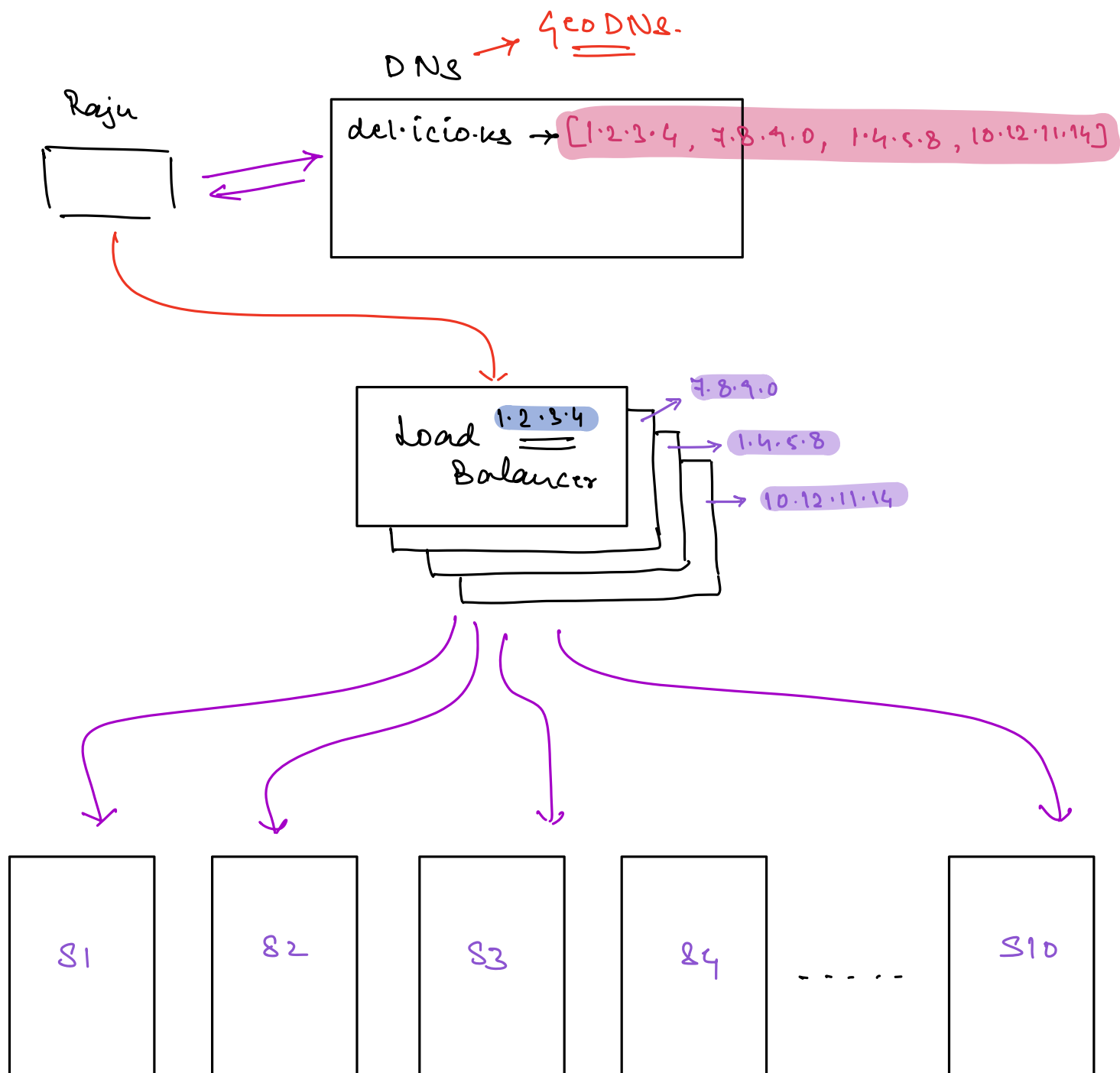


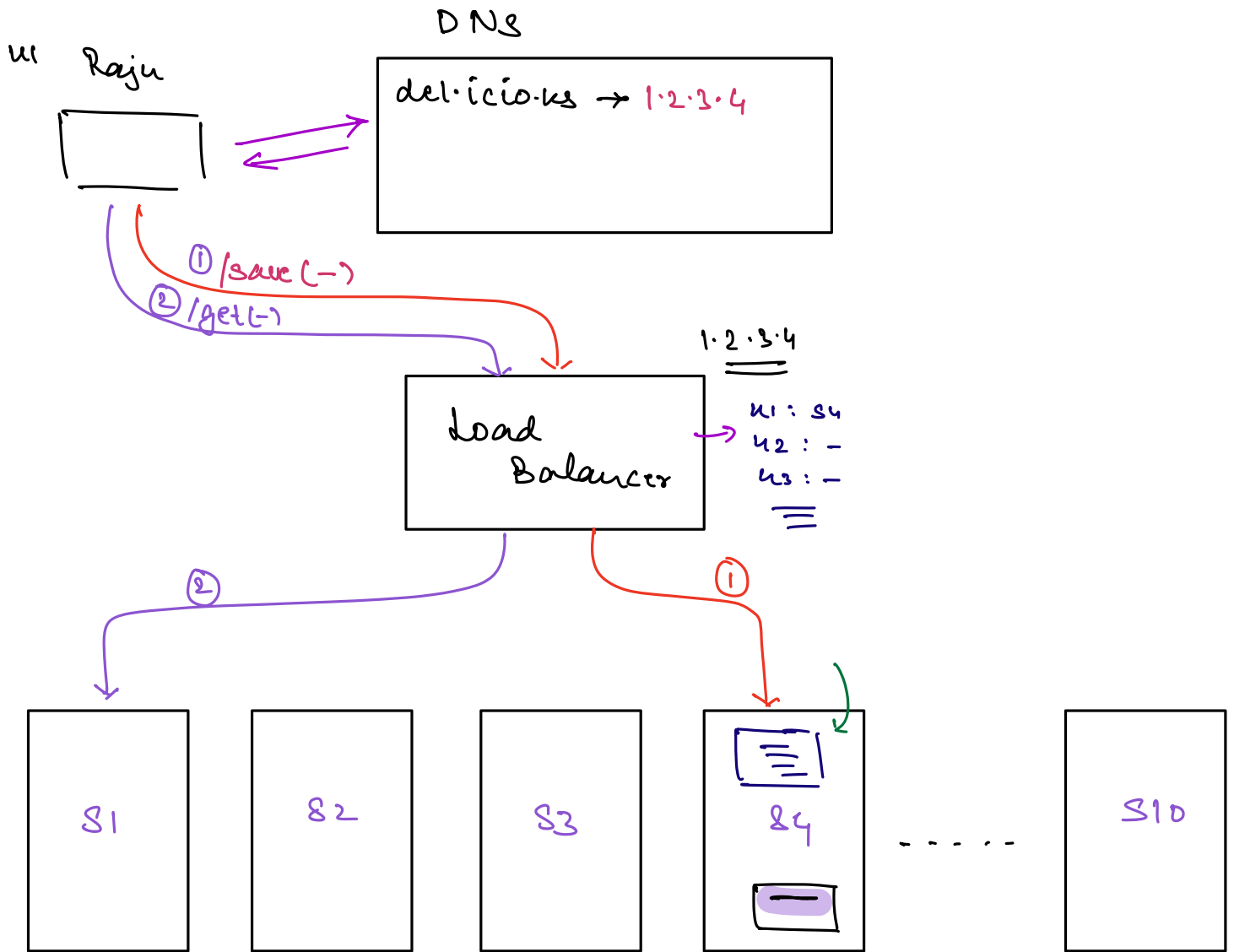
Scaling

- Vertical Scaling
- Horizontal Scaling

Bottleneck.





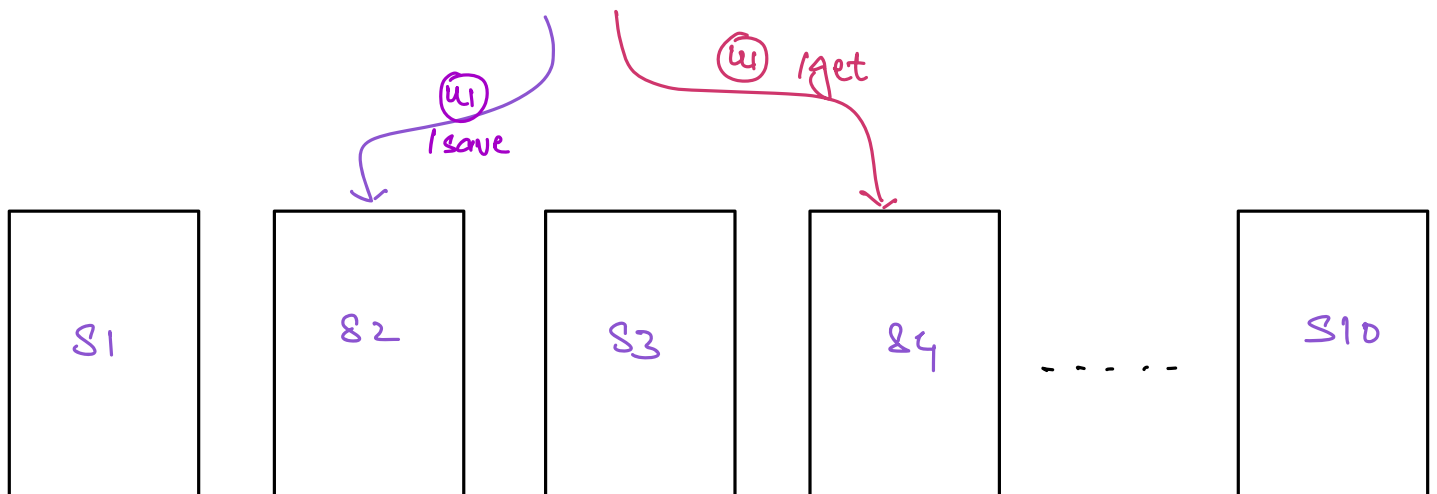
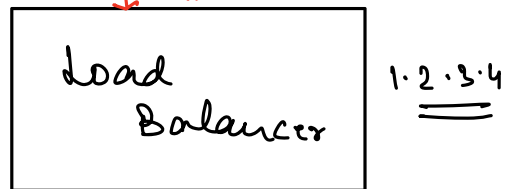
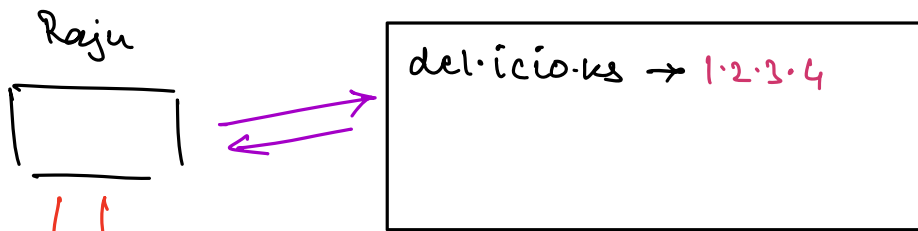


Stateless (vs) Stateful Load Balancing

Tight Coupling

Replica.

DNS



App Server
DB Server

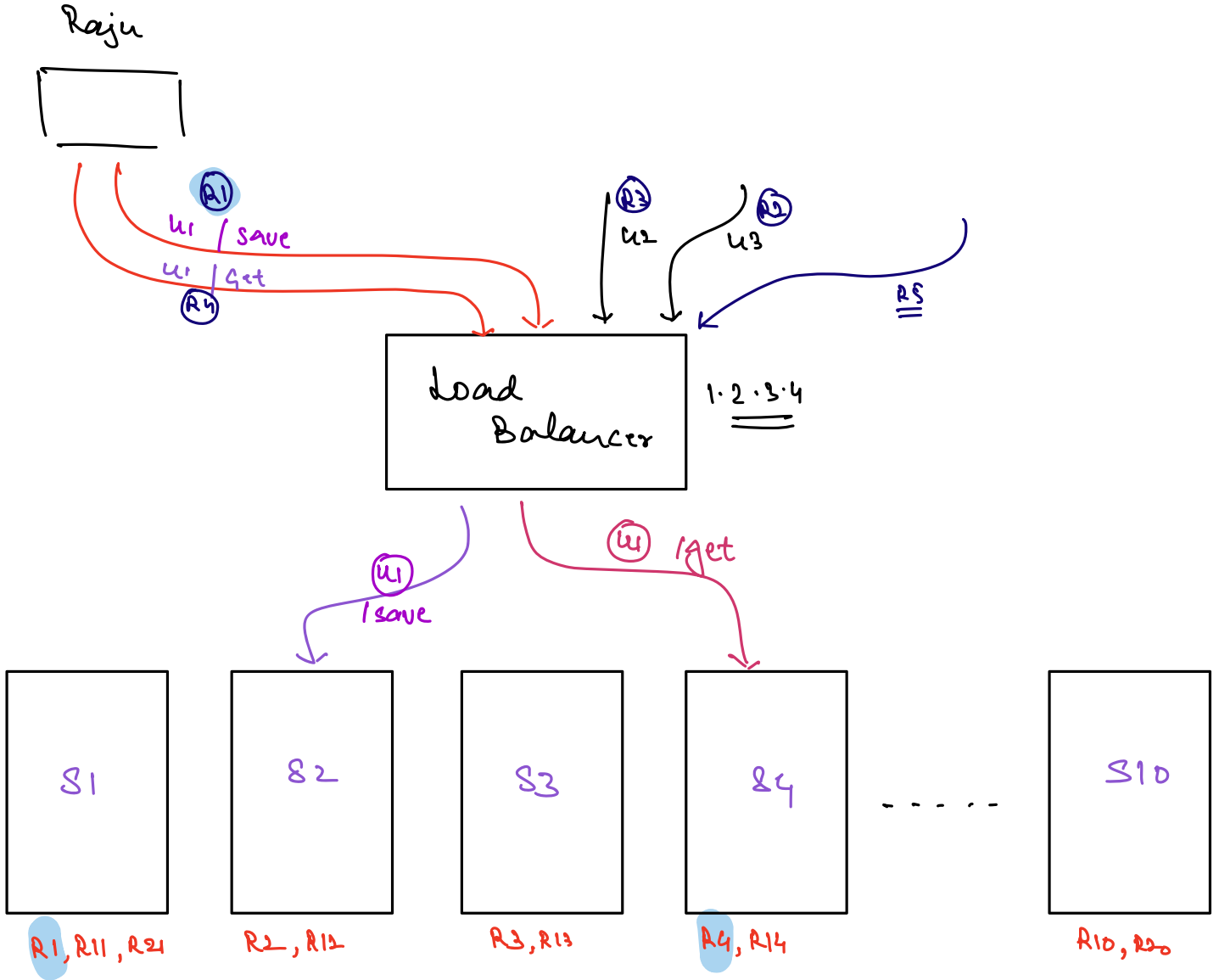
Replicas.

↓
Replication
factor.

Load Balancing Algorithms.

→ Random.

→ Round Robin.

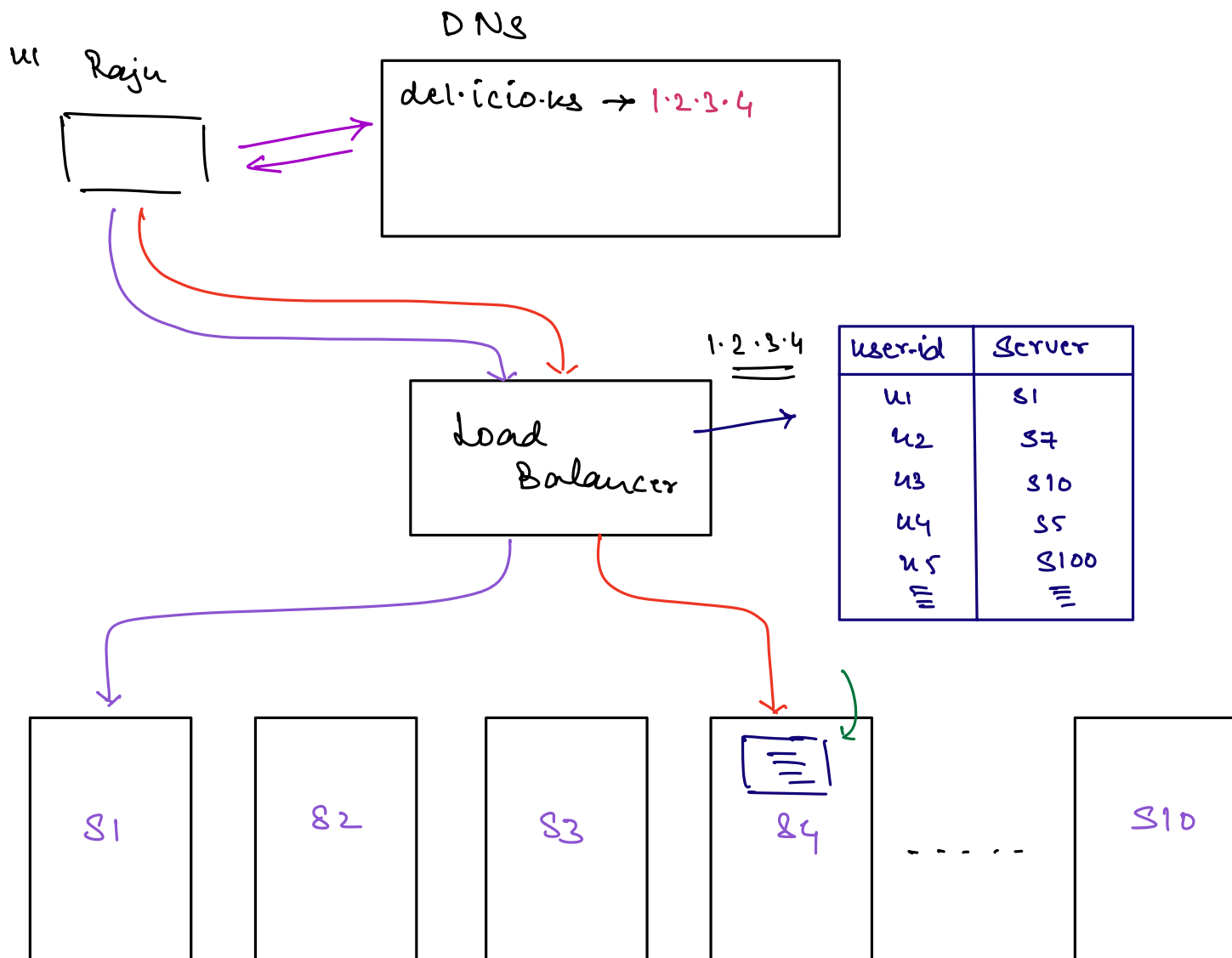


→ Stateless.

→ Least Connection First
↳ Stateless.

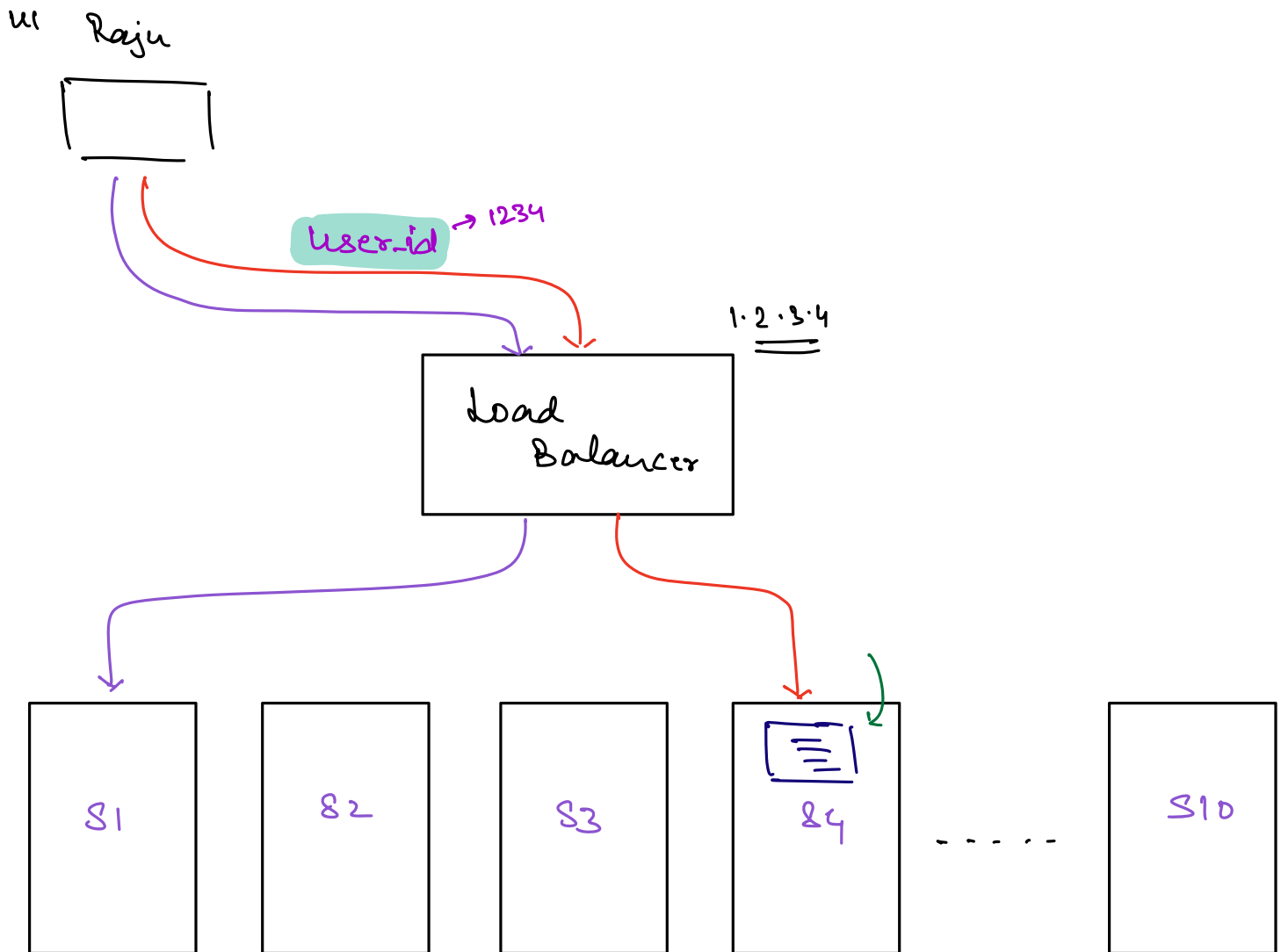
→ Least Response Time.
↳ Stateless.

Stateful Load Balancing Algorithms.



Modulo Based Algorithm.

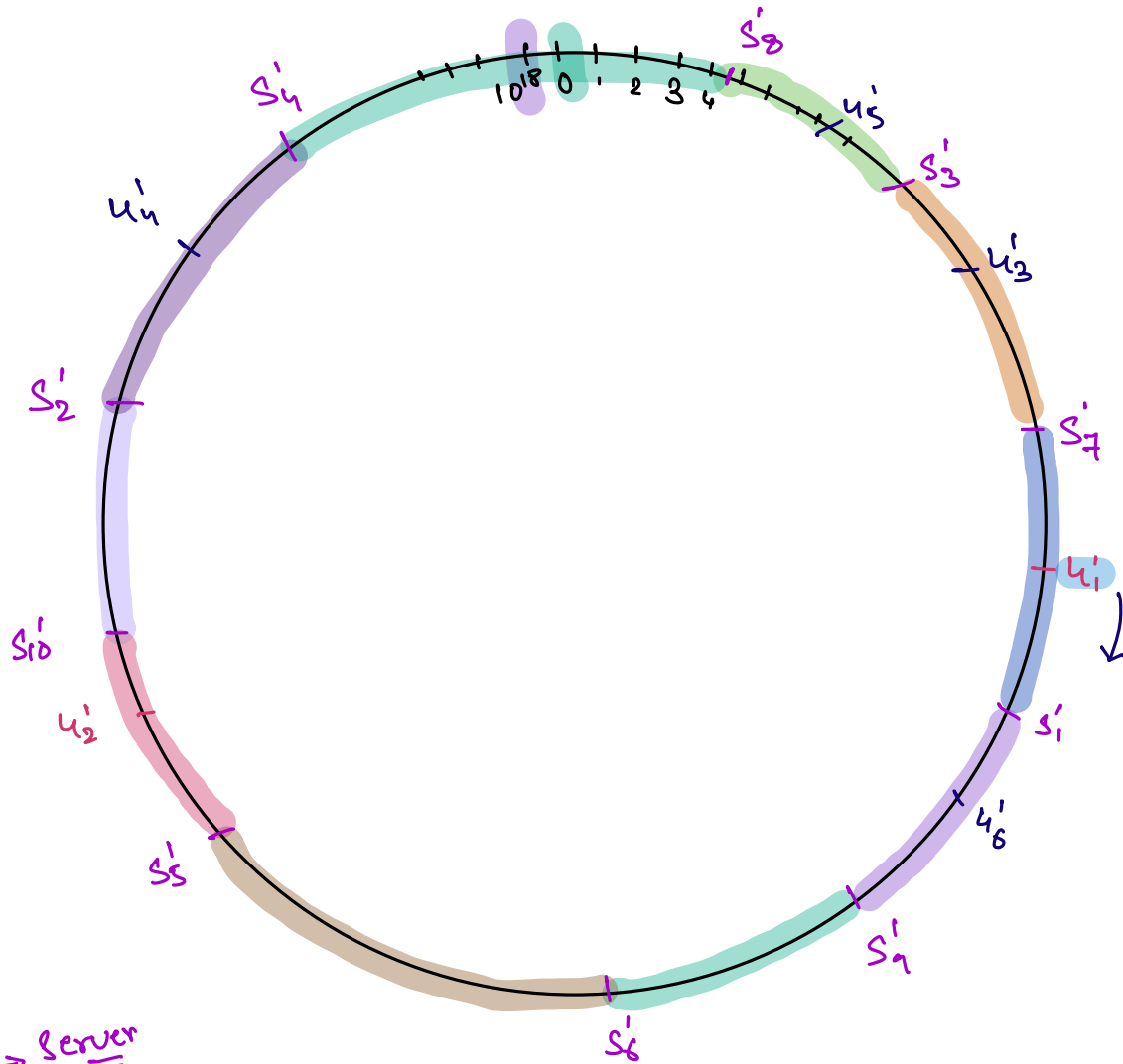
↳ Stateful



$x \% y \Rightarrow$ Remainder when we divide x by y
↳ Modulo Operator.

CONSISTENT HASHING.

↳ Stateful



$H_s \rightarrow$ server
↳ Hash fun

$[0, 10^8]$

$$H_s(s_1) = s'_1$$

$$H_s(s_2) = s'_2$$

$$H_s(s_3) = s'_3$$

⋮

$$H_s(s_{10}) = s'_{10}$$

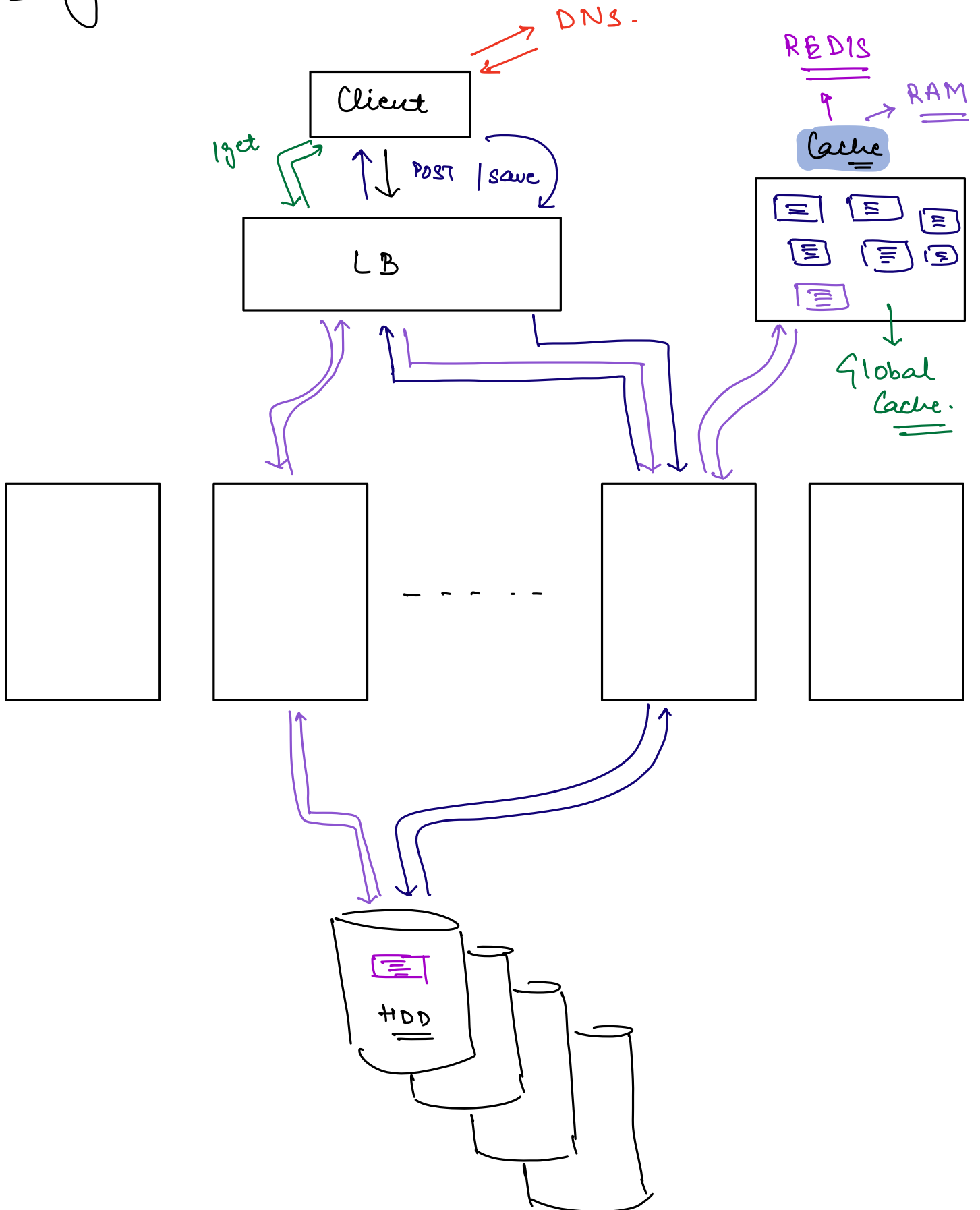
$H_u \rightarrow$ user-id

↳ Other Hash fun

$[0, 10^8]$

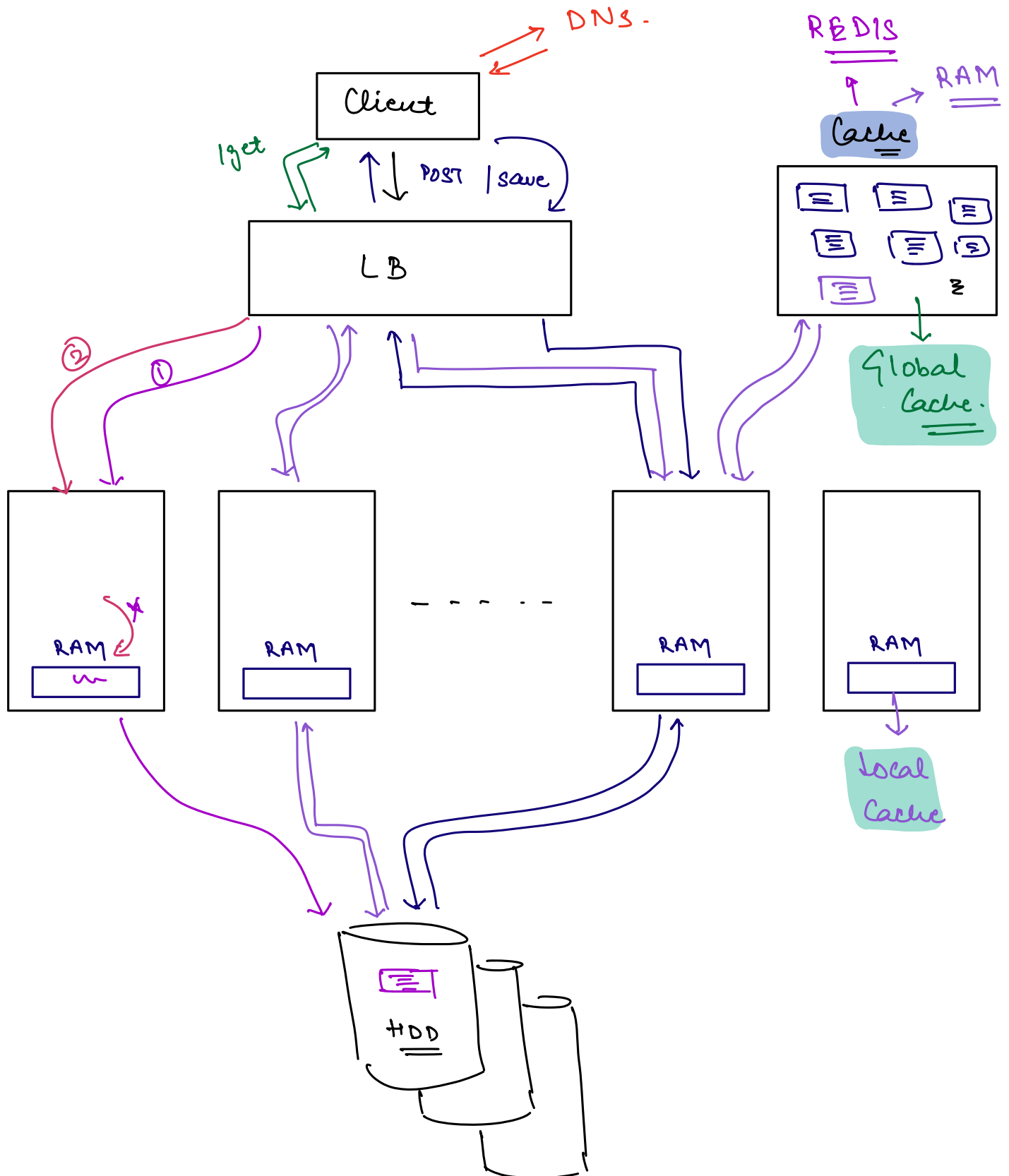
$$H_u(u_1) = u'_1$$

Caching



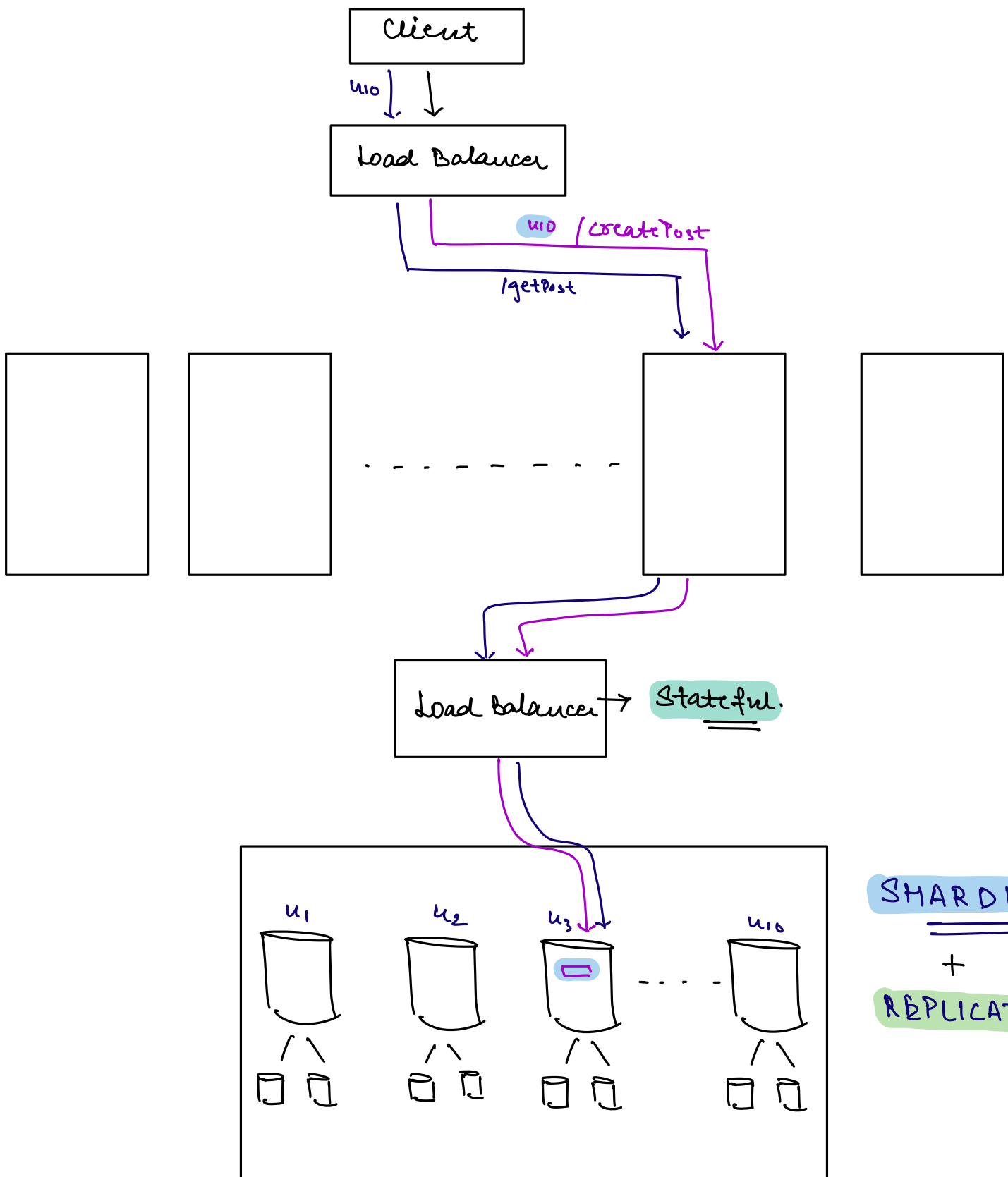
Caching

→ Process of storing copies of most frequently accessed data near by so that this data can be accessed much faster than DB access.



CDN

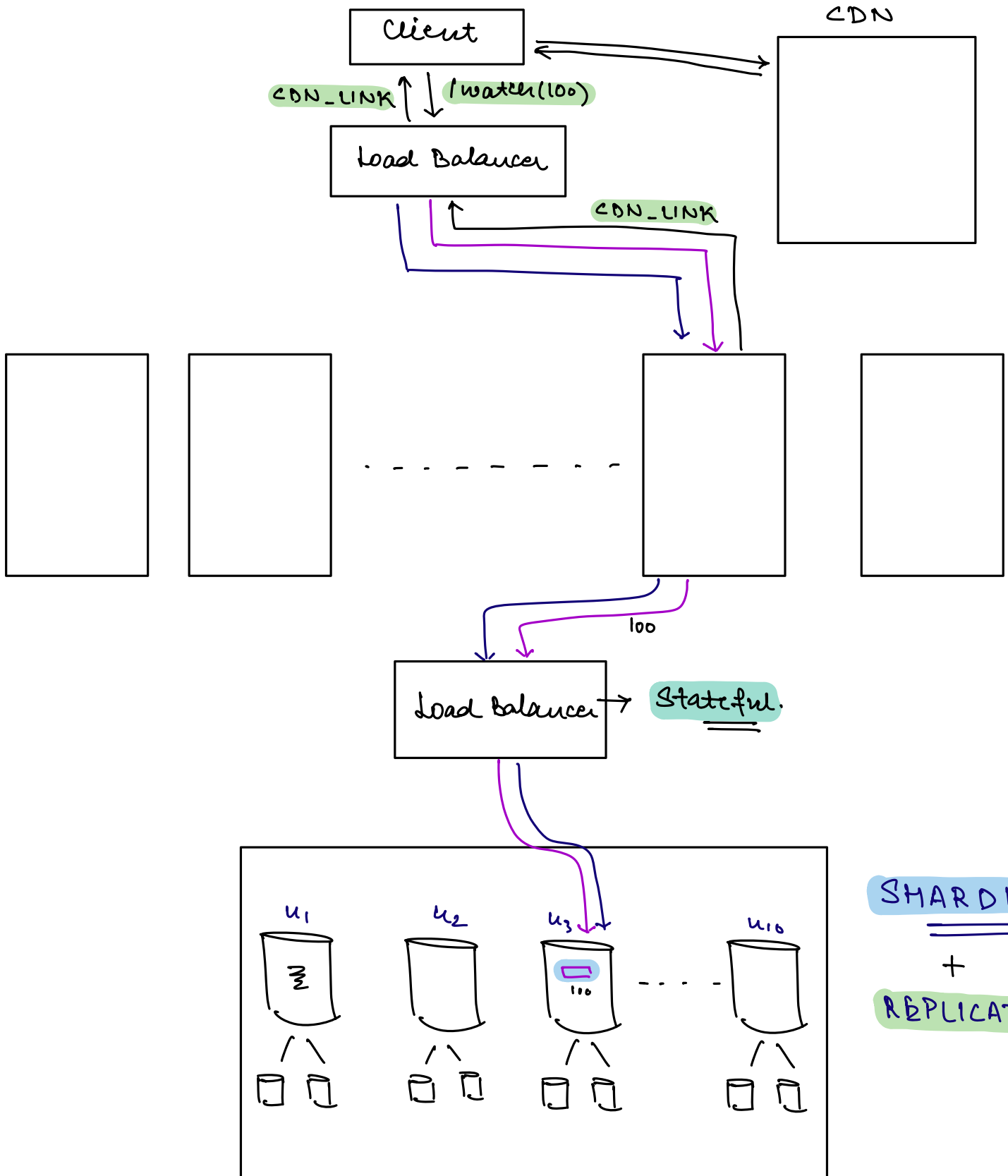
→ Content Delivery Network.

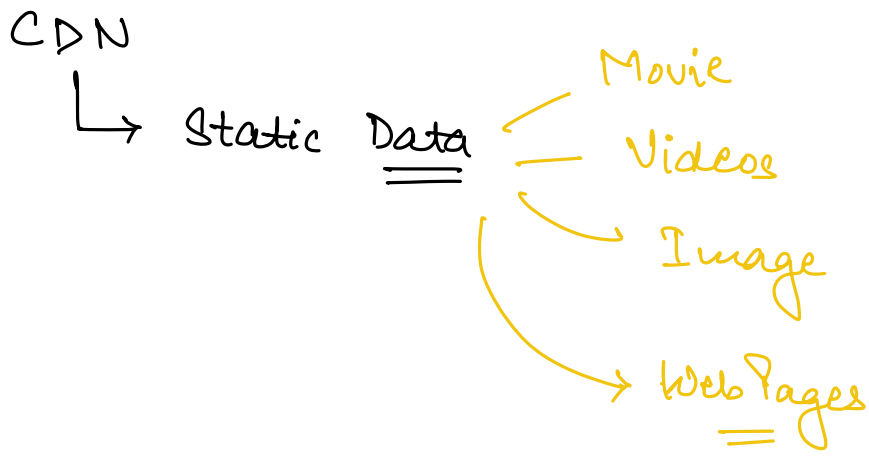


CDN

- Akamai
- Cloudflare
- Cloudfront

Geo Nodes



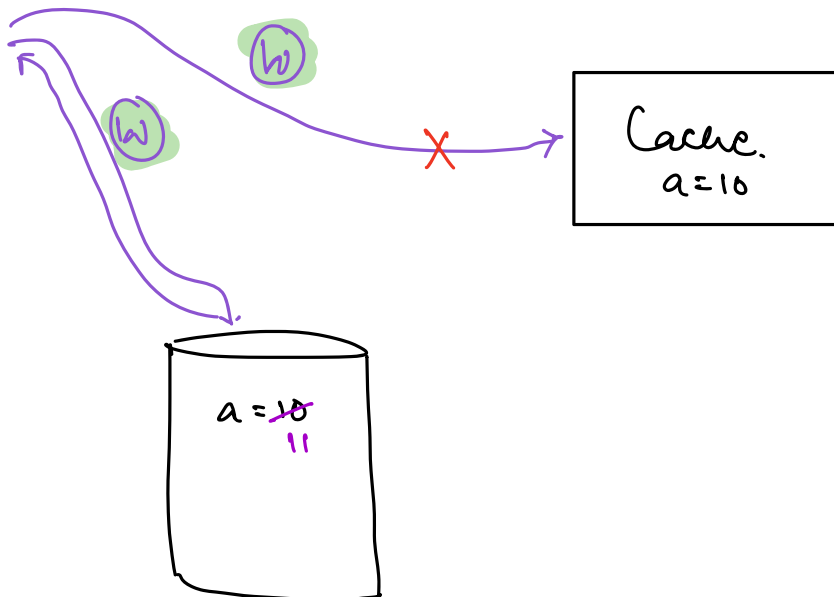


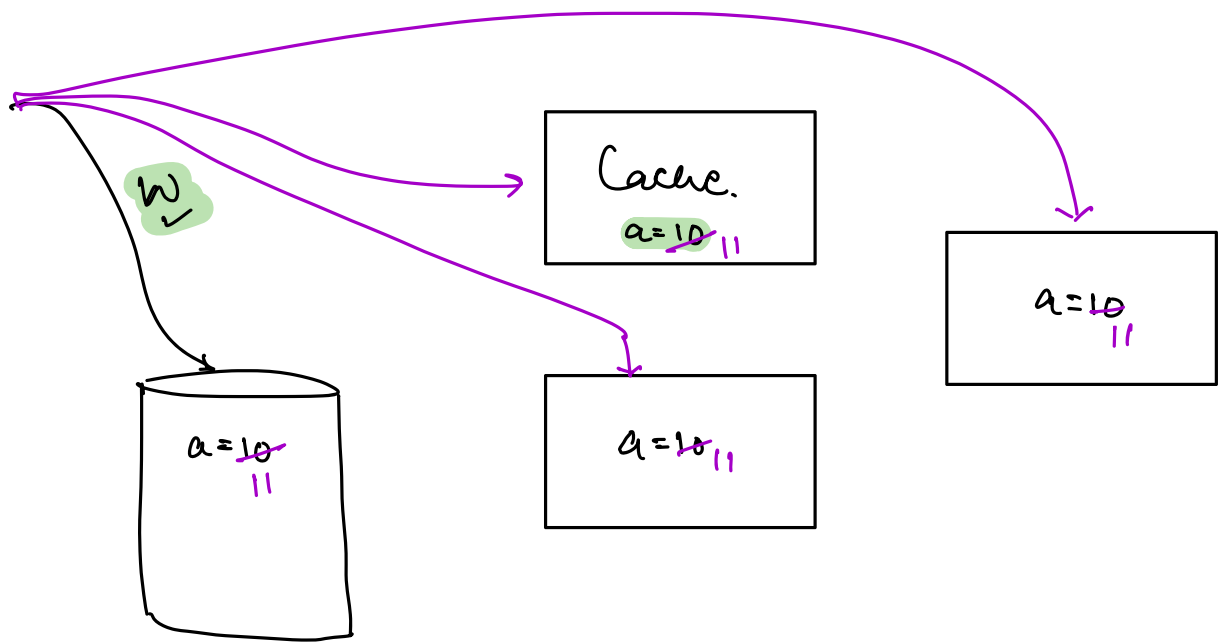
OpenConnect

↳ CDN by Netflix.

Synchronous Update → Strong Consistency =
→ Slow

Asynchronous Update → Eventual Consistency





Cache Eviction

- LRU : Least Recently Used
- FIFO : First In, First Out.
- LFU : Least Frequently Used.
- LIFO : Last In, First Out.
- MRU : Most Recently Used

Cache Invalidation

TTL : Time To Live.

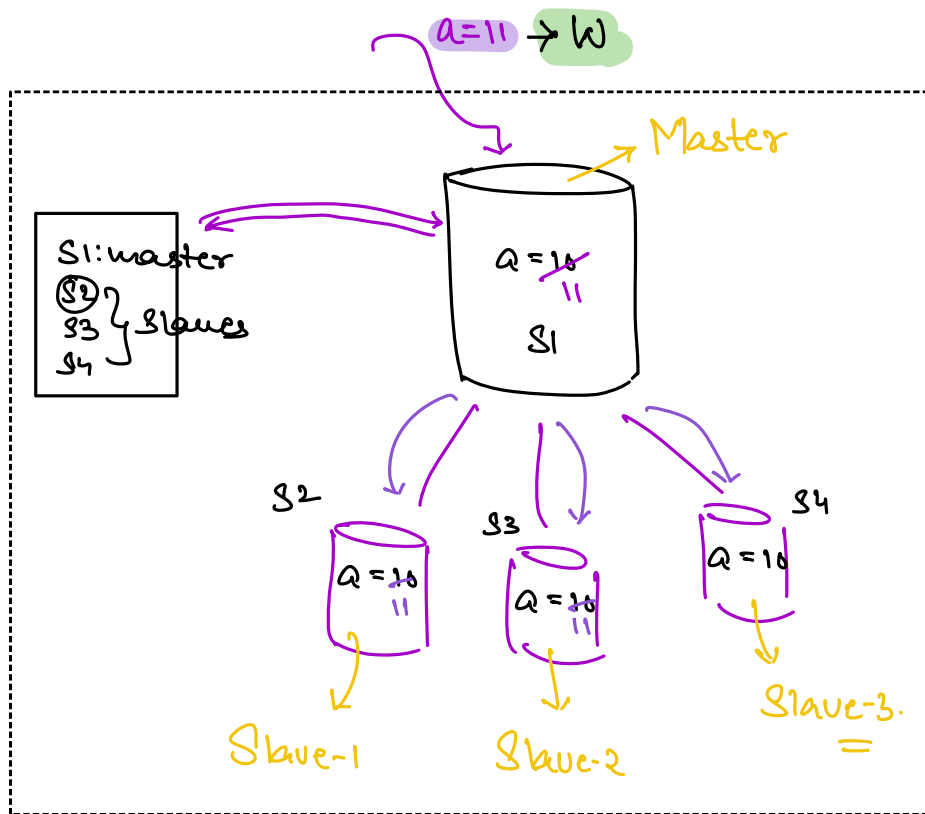
1 min

Cric Buzz

↳ REDIS.

Replication

↳ distribute the read traffic.
(Read throughput ↑)



Master-Slave /
Leader-Follower
Architecture

Write: Only goes to Master.

Strong Consistency

↳ Sync Updates

Eventual Consistency

↳ ASync Updates.

Databases.

↳ Relational or Non-Relational

⇓

SQL

⇓

NoSQL